Hayeon Jeong

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RESEARCH INTERESTS

Visual Generative Models, Post-training for Generative Models, Mechanistic Interpretability, AI Safety/Reliability.

EDUCATION

Yonsei University — *Seoul, Korea*
M.S. in the School of Integrated Technology — *Mar. 2025 – Feb. 2027 (Expected)*

- GPA 4.16/4.30

University of California, Berkeley — *Berkeley, CA, USA*
Visiting Undergraduate Student — *Jan. 2023 – May 2023*

Soongsil University — *Seoul, Korea*
B.S. in Electronic and Information Engineering — *Mar. 2020 – Feb. 2025*

- GPA 4.11/4.50 (Magna Cum Laude)

PUBLICATIONS

1. **Hayeon Jeong** and Jong-Seok Lee (2025). "Dominating vs. Dominated: Concept-Level Generative Collapse in Diffusion Models". arXiv preprint; under review at *ACCV 2026*.
2. Sohee Park, **Hayeon Jeong**, and Daeseon Choi (2025). "Infrared Thermal-Guided Adversarial Patch Defense for Robust Visible Person Detectors". In: *IEEE AVSS 2025*.

RESEARCH EXPERIENCE

UCLA — *Los Angeles, CA, USA*
Visiting Researcher — *May 2026 – Oct. 2026 (Expected)*

- Part of the CML Lab, advised by Prof. Cho-Jui Hsieh.
- Working on *Flip What Fails* (first author): RL post-training of text-to-image models on long real-user prompts with counterfactual pairs that flip only the requirements a model fails, preserving satisfied ones.

Yonsei University — *Seoul, Korea*
M.S. Researcher — *Mar. 2025 – Feb. 2027 (Expected)*

- Part of the MCML Lab, advised by Prof. Jong-Seok Lee.
- Formalized concept dominance in multi-concept diffusion generation and built DominanceBench; showed via controlled fine-tuning that low training-image diversity drives it, and found through head ablation that, unlike memorization, it cannot be removed by ablating any small set of heads **(under review, ACCV '26)**.

Soongsil University — *Seoul, Korea*
Undergraduate Research Intern — *Jun. 2023 – Aug. 2024*

- Part of the AI Safety Center, advised by Prof. Daeseon Choi.
- Trained YOLOv5/YOLOv8 person detectors and ran the full physical-world experiment (printed patches, 20 participants, paired RGB–thermal capture) for a thermal-guided adversarial-patch defense **(AVSS '25)**.
- Built an on-device privacy assistant that summarizes consent forms with LLaMA-3 8B Instruct; key-phrase marking raised precision from 77.3% to 98.8% **(Best Paper Award, KIISC Chungcheong Conference)**.

SELECTED PROJECTS

- **Border Conditions in Adversarial Patch Attacks.** Showed on YOLOv5 that patch border color, thickness, and interpolation shift hiding-attack success rates by up to 9.8 points (4.5 on average); gray 4–8 px borders help most (2nd Place, Undergraduate Thesis Competition, Soongsil University). (project page)
- **Phonetic-Aware Encoder Tuning for L2 Korean Speech Recognition.** Automatic speech recognition (ASR) framework robust to phonetic disruption in non-native Korean speech via a LoRA-adapted Whisper encoder with auxiliary connectionist temporal classification (CTC) supervision. (project page)
- **Lightweight DenseNet.** EMNIST classifier with a DenseNet variant using 10× fewer parameters than ResNet-50 and half the training time, and 2× faster than DenseNet-121. (project page)

HONORS & AWARDS

- **Best Paper Award (KISA President's Award)**, KIISC Chungcheong Conference, 2024.
- **Professor's Scholarship** (50% tuition, merit-based), School of Electronic and Information Engineering, Soongsil University, 2024.
- **Truth Award** – Professor's Scholarship (25% tuition, merit-based), Soongsil University, 2024.
- Academic Excellence Scholarship (25% tuition, merit-based), Soongsil University, 2021.

ACADEMIC SERVICE & TEACHING

- Reviewer, *IEEE ICASSP 2026*.
- Teaching Assistant, *Advanced Engineering Mathematics*, Yonsei University, Spring 2025.

OTHER EXPERIENCE

- Trainee, SK Hynix Semiconductor Curriculum 2.0, SUNI × SK hynix (Sep. 2023 – Dec. 2023); completed coursework on the semiconductor market, memory devices (DRAM, NAND), CIS, and industry trends.
- Data Annotator, AI Training Data Construction Project, Univa (Jul. 2022 – Sep. 2022); annotated mathematical-equation images to build supervised training data for an OCR recognition model.
- Member, Baird Volunteer Corps, Soongsil University (Oct. 2020 – Feb. 2022); contributed to community safety through a CCTV installation project with the local police station and youth education programs.
- Mentor, KT Mentoring Program (Oct. 2020 – Mar. 2022); developed a 6-month Math/English curriculum for middle school students.

SKILLS

- **Languages:** Korean (native), English (professional working proficiency).
- **Programming:** Python, Java, C/C++.
- **Frameworks:** PyTorch, Hugging Face (Diffusers/Transformers).
- **Tools:** Git, LaTeX.